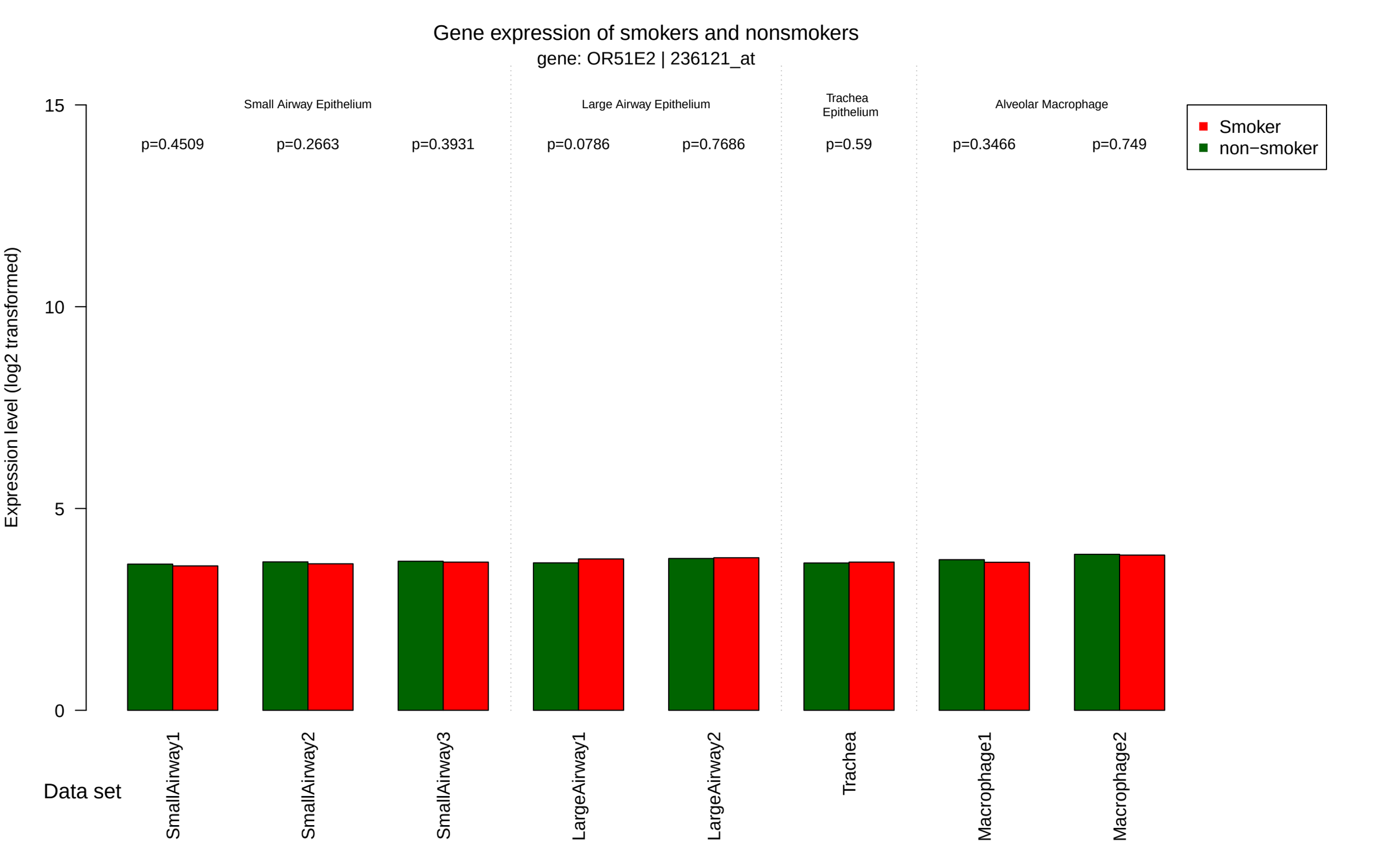
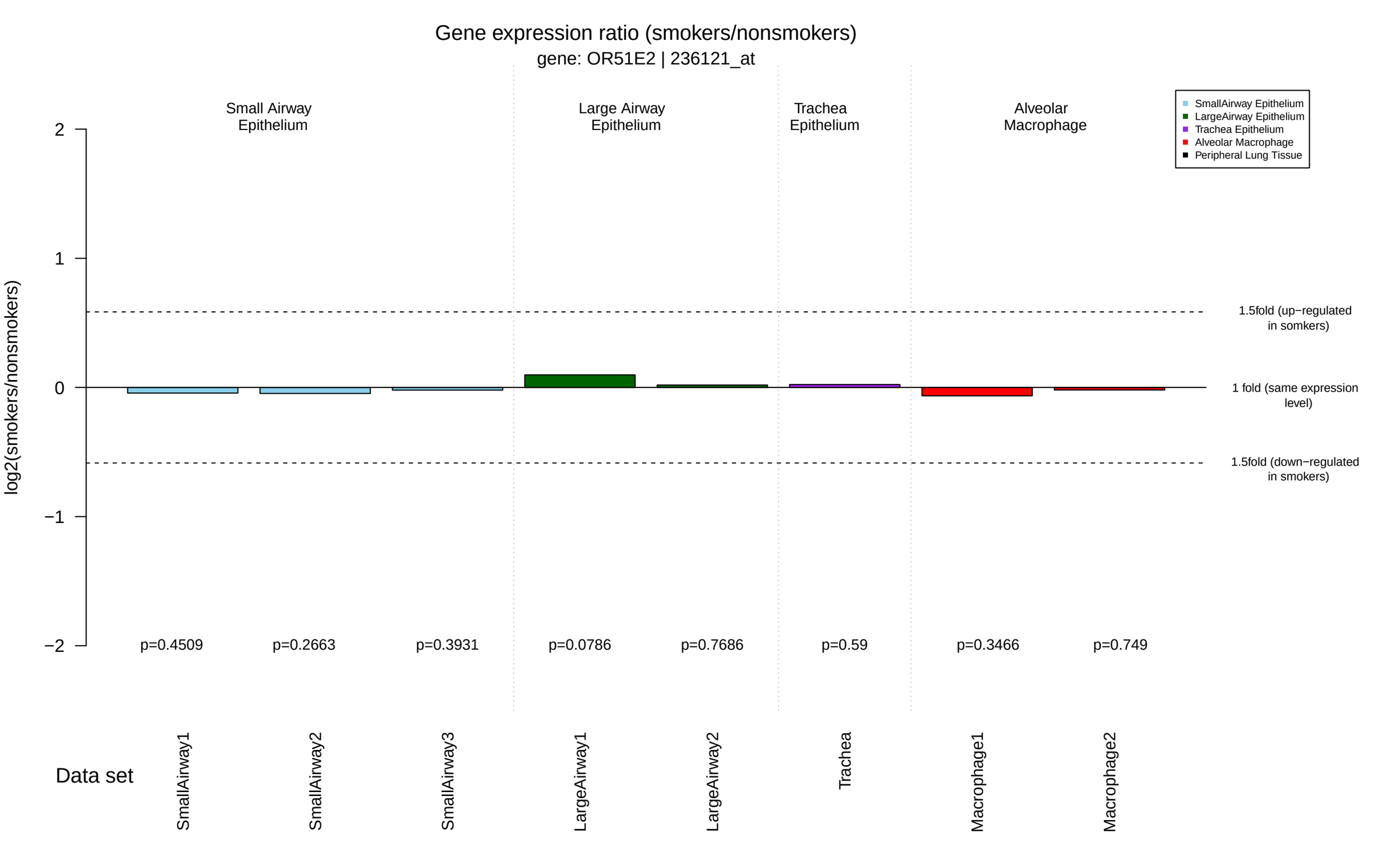
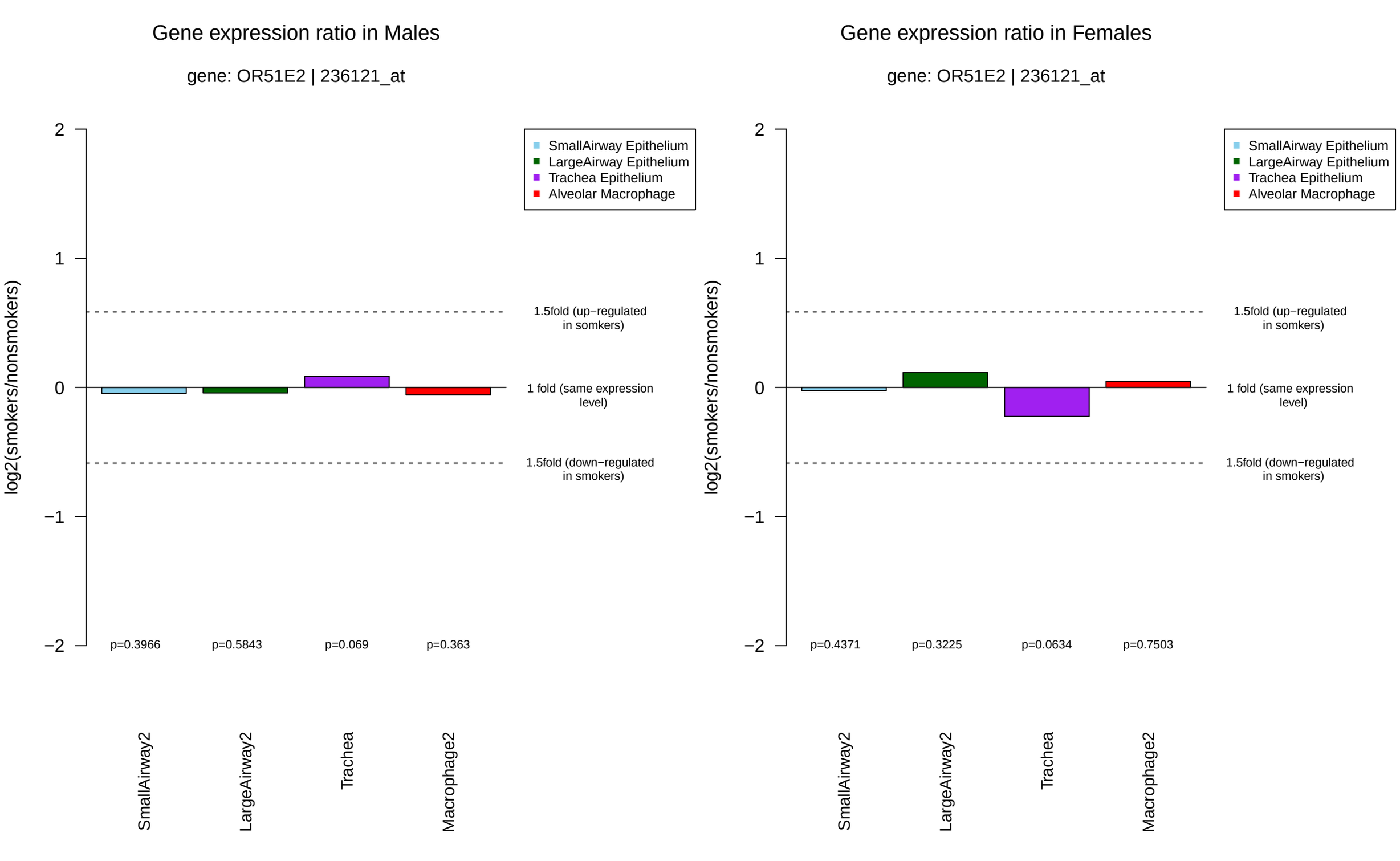
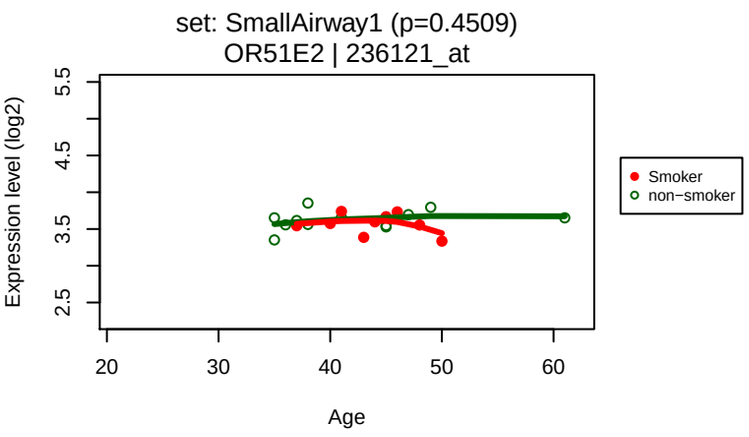
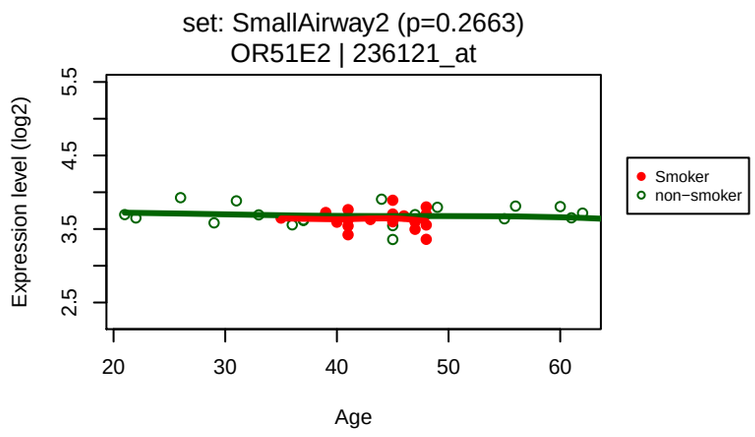
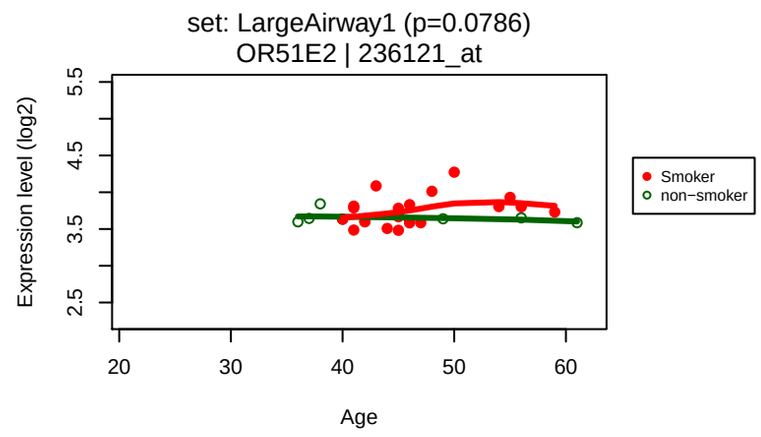

Gene expression of smokers and nonsmokers
gene: OR51E2 | 236121_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4509
p=0.2663
p=0.3931
p=0.0786
p=0.7686
p=0.59
p=0.3466
p=0.749
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

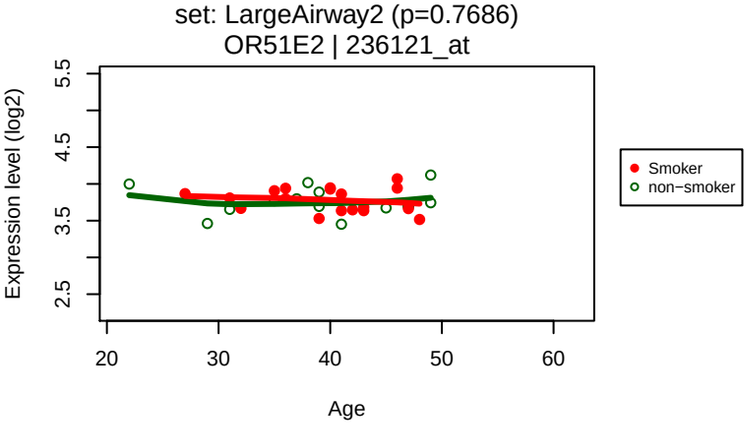

Gene expression ratio (smokers/nonsmokers)
gene: OR51E2 | 236121_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4509
p=0.2663
p=0.3931
p=0.0786
p=0.7686
p=0.59
p=0.3466
p=0.749
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

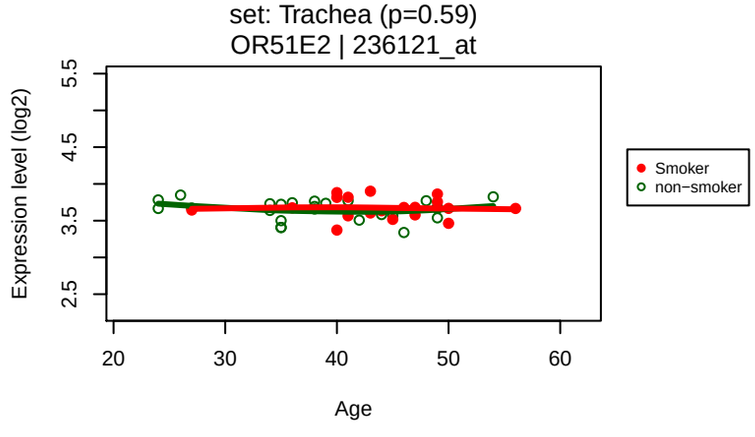

Gene expression ratio in Males
gene: OR51E2 | 236121_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3966
p=0.5843
p=0.069
p=0.363
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OR51E2 | 236121_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4371
p=0.3225
p=0.0634
p=0.7503
SmallAirway2
LargeAirway2
Trachea
Macrophage2

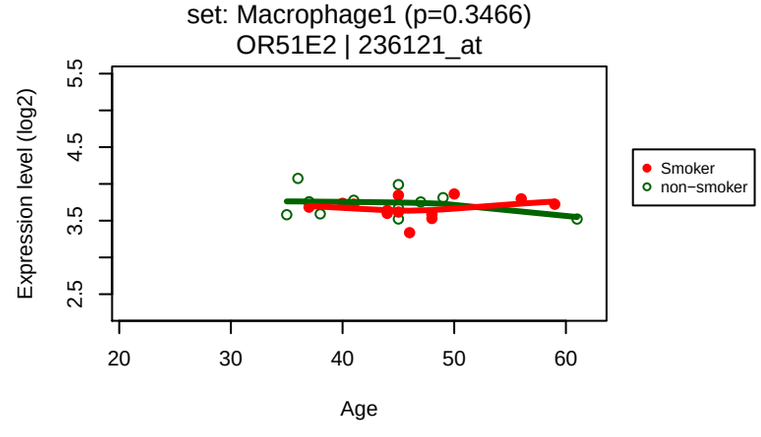

set: SmallAirway1 (p=0.4509)
OR51E2 | 236121_at
Expression level (log2)
Age
Smoker
non−smoker

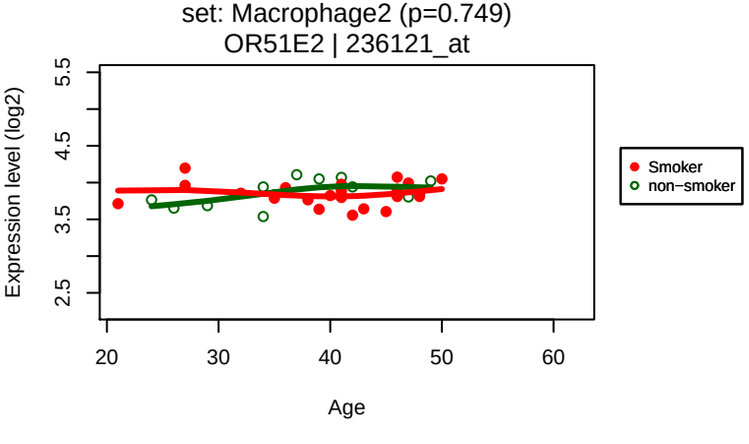

set: SmallAirway2 (p=0.2663)
OR51E2 | 236121_at
Expression level (log2)
Age
Smoker
non−smoker

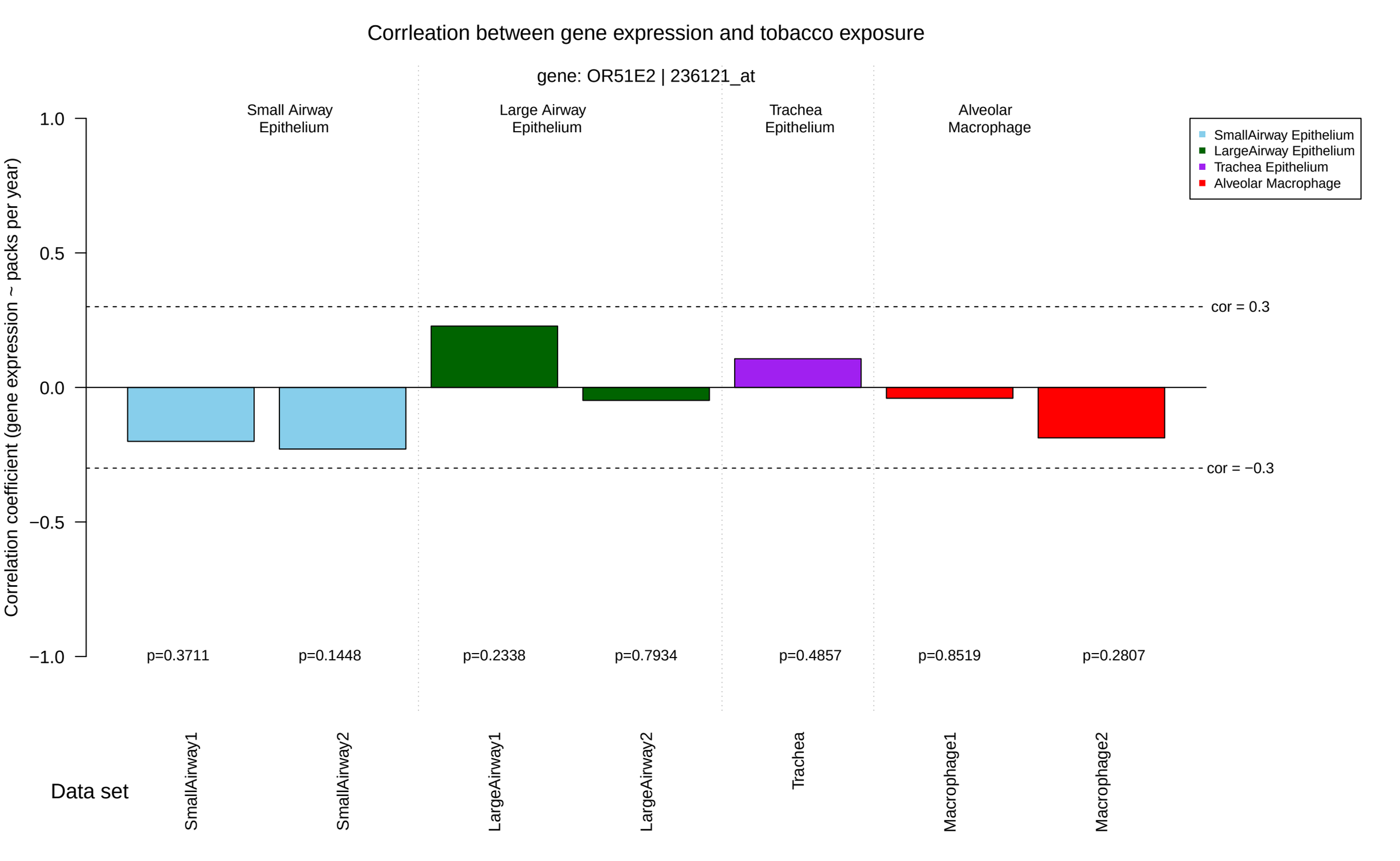

set: LargeAirway1 (p=0.0786)
OR51E2 | 236121_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7686)
OR51E2 | 236121_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.59)
OR51E2 | 236121_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3466)
OR51E2 | 236121_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.749)
OR51E2 | 236121_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OR51E2 | 236121_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3711
p=0.1448
p=0.2338
p=0.7934
p=0.4857
p=0.8519
p=0.2807
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2